

Privacy and Personal Space: Addressing Interactions and Interaction Data as a Privacy Concern

Lauren E. Buck*
Vanderbilt University

Bobby Bodenheimer†
Vanderbilt University

ABSTRACT

The privacy and security of personal data has been at the forefront of public concern for some time now, and is typically understood in the context of data collected from online interaction (social media, transactions, search engine queries, etc.). The advent of immersive technologies expand data collection beyond what can typically be extracted via online interaction, particularly in terms of the availability of biometric data (eye tracking and gait analysis). However, it has not yet been brought to light that interactions and interaction data will soon be of concern in terms of privacy. We mediate interactions in everyday life through the maintenance of personal space and allow certain individuals and objects into our personal space. We do the same in virtual reality. Our personal space allows us to preserve our feeling of safety, and the way we mediate it shows our biases and preferences. In this work, we take a look at the implications of interaction and the availability of personal data that will bring a host of ethical and privacy concerns.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Security and privacy—Human and societal aspects of security and privacy—Social aspects of security and privacy

1 INTRODUCTION

Virtual reality has become more ubiquitous as a commodity level technology in recent years with Google releasing Cardboard¹, Oculus the Rift and Quest², and HTC the Vive³, among many other systems. At varying degrees, commodity level systems have the ability to track the movements of users – they can track low level biometric data such as the acceleration and position of the headset as well as hand movements through controllers used to interact with virtual environments and eye movements with built in eye tracking technology. This data can be used to understand how users are interacting with the space around them, and can determine the unique characteristics of the individual using the system.

There is a lot that can be learned from data collected from users, and literature shows us that eye tracking [5] and gait analysis [2] are of major concern in the privacy community. Eye tracking provides sensitive information like preferential gaze fixation [8], or what a user chooses to lend their visual attention to. Gait analysis can be used to identify individuals [6] and gait instability that reveals health problems in individuals [3]. The identification of and characteristics of individuals that can be gathered from this type of data allows those who have access to understand virtual reality users on a deeper level that is not typically readily available. But perhaps this data is the tip of the iceberg for what can be extracted from the use of virtual reality technology.

*e-mail: lauren.e.buck.1@vanderbilt.edu

†e-mail: robert.e.bodenheimer@vanderbilt.edu

¹<https://arvr.google.com/cardboard/>

²<https://www.oculus.com/rift/>

³<https://www.vive.com/us/>

We want to particularly address here interaction itself and the data it provides. As research presses forward, it is becoming more clear how users of virtual reality treat the personal space around their bodies. Every individual maintains space around their bodies that they consider their intimate space, and when this space is intruded upon there are varying levels of discomfort that one experiences depending on what interaction is taking place. How people maintain this space is dictated by a variety of different factors such as their gender, age, race, culture, subculture and mental state, to name a few [4]. There is evidence from neuroscience literature that stress increases as the personal space around the body is intruded upon; cortisol levels in the brain increase, electrodermal activity increases, and there is increased activation of the amygdala when personal space is intruded upon [7]. Thus we believe that there are two important characteristics of privacy in immersive virtual environments that have been neglected – we are concerned about the data that can be gathered from the maintenance of personal space around the body as well as the safety of this space for interaction within immersive virtual environments.

It becomes clear that there is an enormous amount of personal data that can be extracted from how users treat the space around themselves, and that the way interaction in virtual environments takes place is important to preserving the privacy and security of the user. Virtual environments are not currently designed to facilitate close interaction within near space that protects the user. It should be taken into account how different factors of immersive environments, such as avatars, affect the personal space boundaries and comfort of users. We wish to shed light on the potential problems that arise from interaction and interaction data and call them to the attention of the virtual reality community in order to spark dialogue on how they might be addressed.

2 THE ROLE OF PERSONAL SPACE IN PRIVACY

Providing your biometric data to virtual reality systems is something that is inevitable - you cannot use a virtual reality system without providing your head movements, and you cannot interact with 3D environments without some form of input, whether it be through button presses or the tracking of controllers and tracking pucks and/or markers. Virtual environments are inherently designed to be engaging and to provide rich experiences to users, and the goal of this technology is to provide interaction that is similar to what takes place in the real world. Our preferences become clear through interaction – we decide what food, clothes, entertainment and people to engage with – and this takes place within our intimate space.

Thus how people engage in interaction with the space around them becomes of great concern. Personal space is the minimum distance that one feels comfortable interacting with an object or person. We only allow interaction with objects and people in this space that we feel comfortable with. Personal space is plastic, meaning that it expands and contracts based on the interaction taking place. It expands when undesirable interaction takes place and contracts when desirable interaction takes place. Thus this space reveals important distinct preferences that each individual maintains. It reveals what items and people we have an affinity for and what items and people we are averse to. It reveals implicit biases and the state of our mental health. How we mediate the space around us can alert anyone with

access to our data to the fact that we specifically enjoy Cherry Coke on a regular basis and that we like cats and playing chess and that we have social anxiety. It can reveal the people that we interact with on a regular basis including family and friends.

We have the ability to quantitatively measure how users of immersive virtual environments treat the personal space around themselves [1], and this measurement has shown users of immersive environments to treat objects and agents similar to how objects and people are treated with regard to personal space in the real world. This creates budding concern for how we treat such personal information. If potential employers, merchants, and healthcare providers have this information it may be possible to discriminate based on our personal preferences and psychological profile. Online organizations may have the ability to advertise products and media that become detrimental to one's well being. There are a plethora of other problems that might arise from the availability of this data.

An additional problem to be addressed is how interaction with personal space is mediated in immersive virtual environments in order to protect the privacy and safety of the user. It is common knowledge that the anonymity provided by online outlets spurs interactions that would not typically take place in the real world. Transferred to a 3D immersive virtual environment, the implications of this behavioral shift become poignant. With the ability of users to easily feel embodied and present in immersive virtual environments, interaction takes place on a highly personal level. Interaction can also take place in scenarios that are not restricted to the limitations of the real world. For example, users can have avatars that have an appearance that is distinctly different than their real life appearance. This information coupled with findings from neuroscience literature that stress increases as interaction takes place within personal space demonstrate the importance of conveying comfortable interactions within personal space. It must be understood that interaction, especially within the space that we consider intimate, does not need to violate the interaction preferences of the user.

This brings into question how researchers and those tasked with the responsibility of data collection and management address the availability of this type of data. This type of data can be crucial in understanding how to preserve safe personal space in immersive virtual environments. It can be beneficial in treating those with different mental disorders by creating new avenues for therapies where they have not previously been available, and it can also allow us to have a greater grasp on the psychological problems unsolved by traditional research methodologies. There are many positive and negative applications of this type of data, and the virtual reality and privacy communities at large must be aware of the looming privacy and security problems surrounding the maintenance of personal space in these types of environments.

3 CONCLUSIONS AND FUTURE WORK

How users of immersive virtual environments allocate and maintain personal space opens up new territory in terms of data availability and privacy. The purpose of this talk is to draw attention to these issues and generate conversation about how this information is to be used by researchers and the virtual reality community at large. With the increasing ubiquity of virtual reality technology, the availability of this type of data is an unavoidable reality.

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